Additional file 1

acrf: a chamber-aware reference framework for cross-cohort integration of atrial transcriptomes

Supplementary Figures S1–S8

Combined searchable PDF of Supplementary Figures S1–S8 for the BMC Bioinformatics Research Article. Internal labels may retain Figure S1–Figure S8 nomenclature.

Table of contents

Figure S1. Cohort overview for public atrial resources used in the study.

Figure S2. Donor pairing and sample-attribute structure for paired-chamber and duration cohorts.

Figure S3. Full LA single-nucleus family and program localization matrices for GSE255612.

Figure S4. Per-cohort principal-component overview of bulk atrial transcriptomes.

Figure S5. GSE238242 RNA-side and mural-cell sensitivity views (non-main-text evidence).

Figure S6. Chamber-contribution views for AF remodeling on the frozen family topology.

Figure S7. F4 constituent-program example supporting family-level interpretation.

Figure S8. Full external AF gene-set overlap matrix against the frozen acrf-eligible background.

Figure legends

Figure S1. Cohort overview for public atrial resources used in the study.

Figure S2. Donor pairing and sample-attribute structure for paired-chamber and duration cohorts.

Figure S3. Full LA single-nucleus family and program localization matrices for GSE255612.

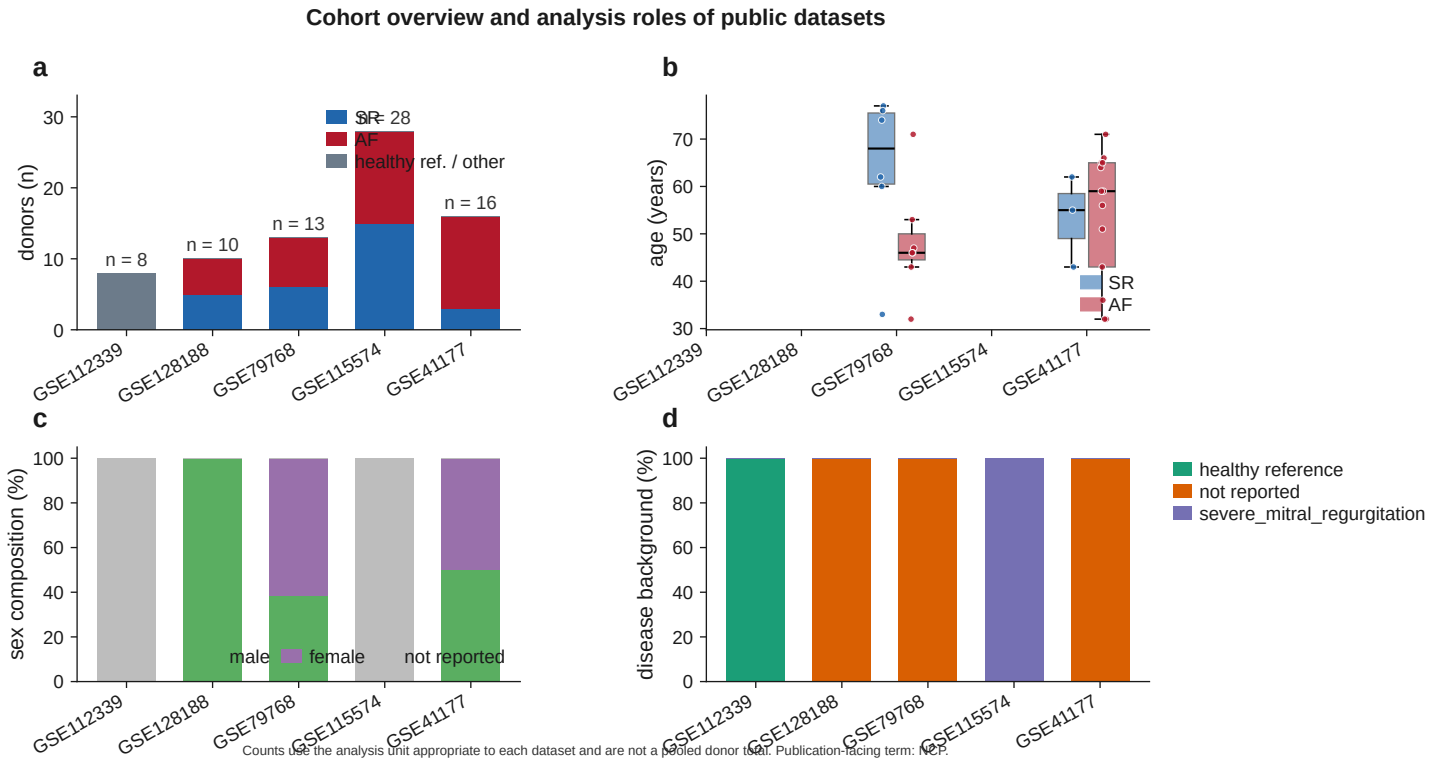
Figure S4. Per-cohort principal-component overview of bulk atrial transcriptomes.

Figure S5. GSE238242 RNA-side and mural-cell sensitivity views (non-main-text evidence).

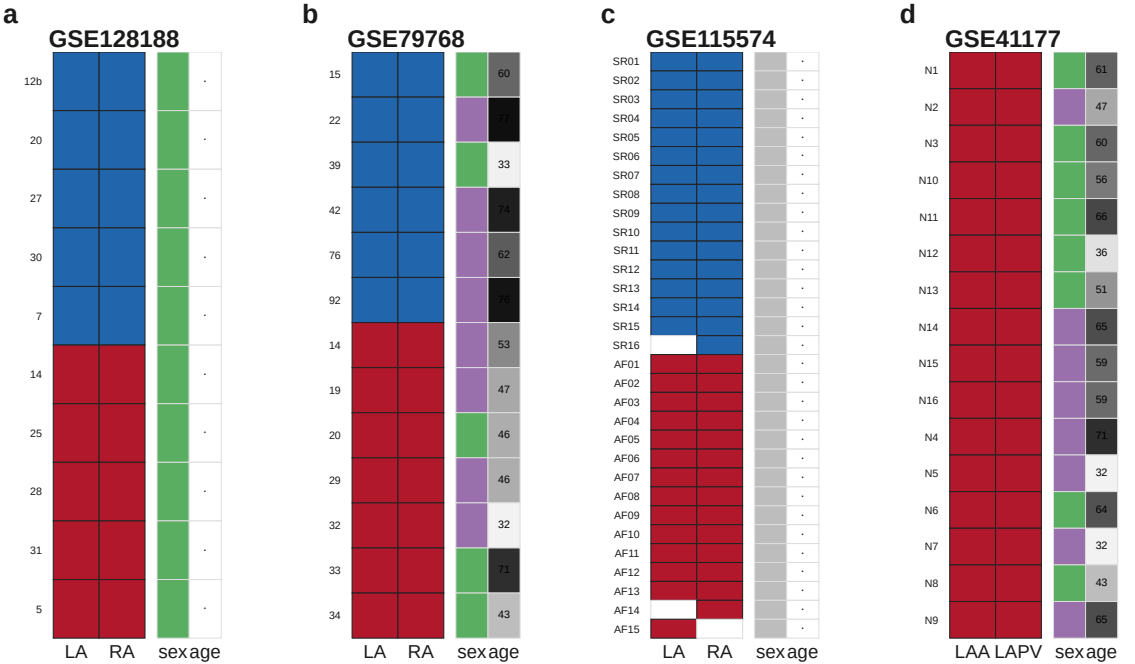
Figure S6. Chamber-contribution views for AF remodeling on the frozen family topology.

Figure S7. F4 constituent-program example supporting family-level interpretation.

Figure S8. Full external AF gene-set overlap matrix against the frozen acrf-eligible background.


Cohort overview and analysis roles of public datasets
a
donors (n)
SR
AF
healthy ref. / other
n = 8
n = 10
n = 13
n = 28
n = 16
GSE112339
GSE128188
GSE79768
GSE115574
GSE41177
b
age (years)
SR
AF
c
sex composition (%)
male
female
not reported
d
disease background (%)
healthy reference
not reported
severe_mitral_regurgitation
Counts use the analysis unit appropriate to each dataset and are not a pooled donor total. Publication-facing term: NCP.

Donor pairing and sample-attribute structure



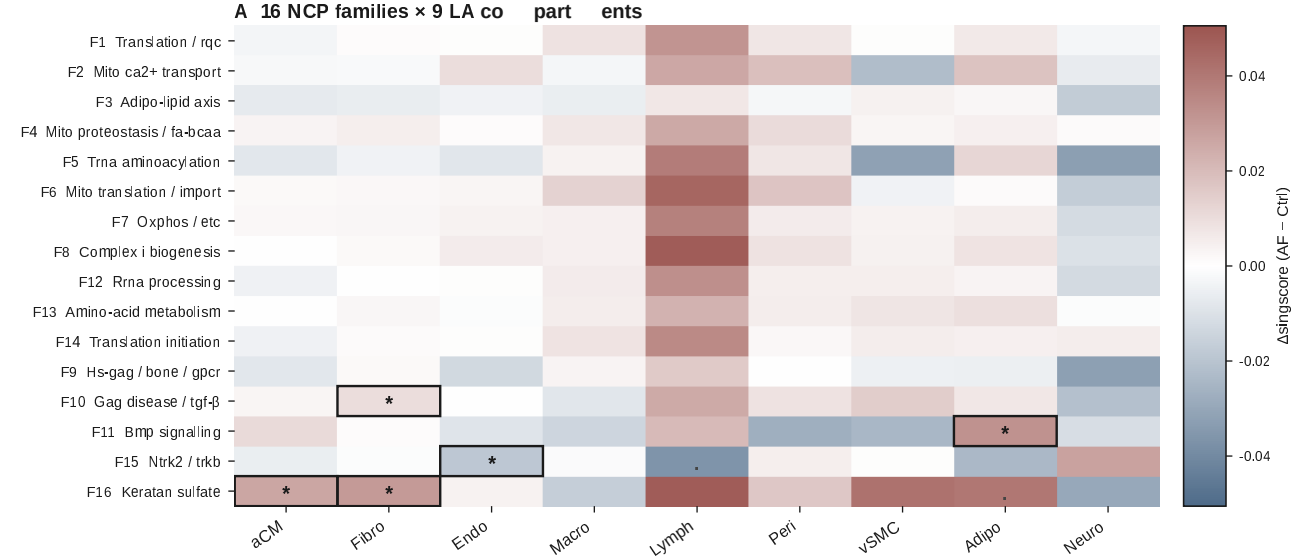
GSE41177 SR samples document cohort structure only; they were not used as duration anchors in the final Figure 4 inference or final Figure 4D panels. Datasets are not a pooled donor denominator.

SR sample present | AF sample present | missing | male | female | not reported

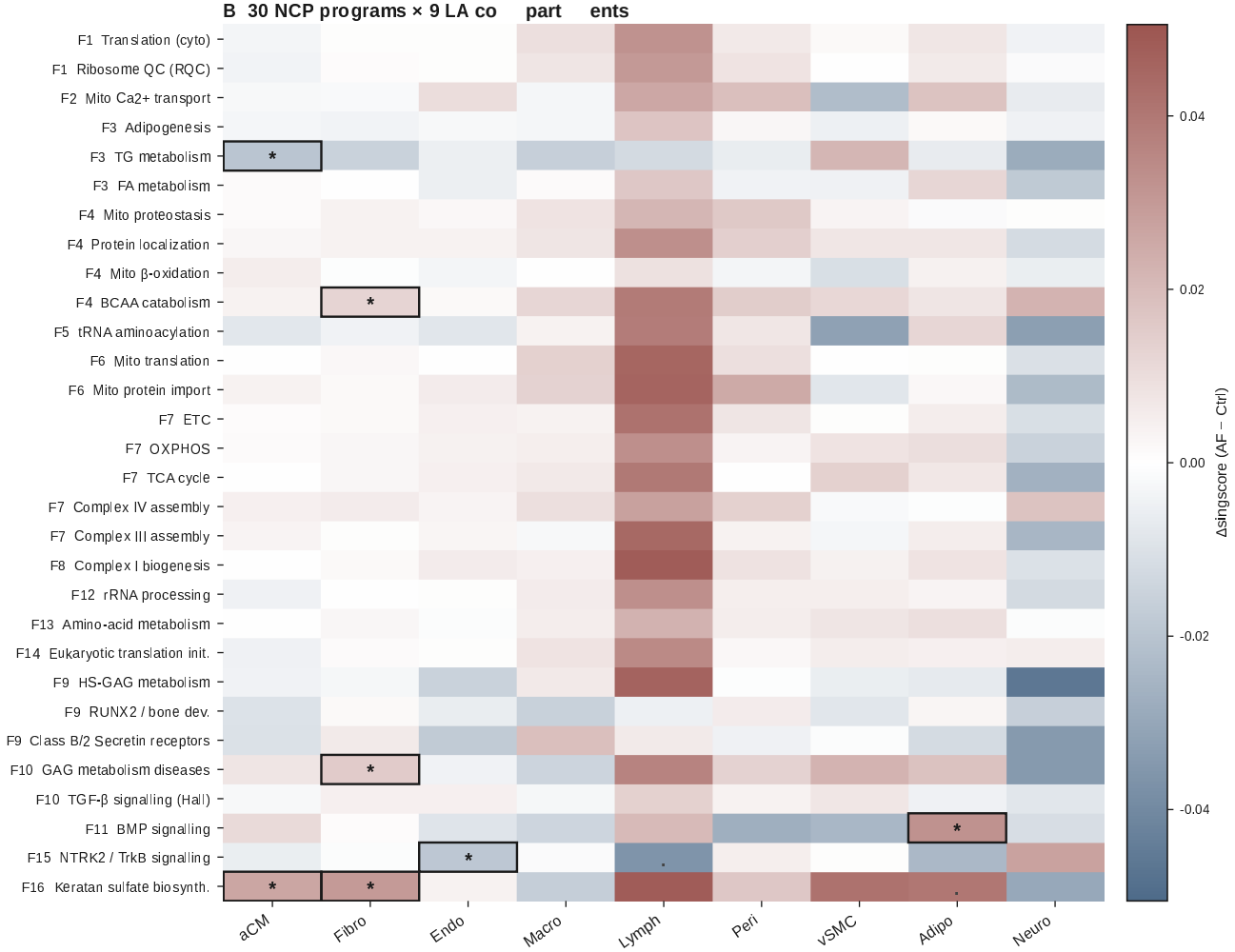
Full LA single-nucleus family and program localization

NCP · GSE255612 LA · exact-reference singscore lineage · Fidelity F10 (fibroblast); Relay F11 (adipocyte), F15 (endothelial only); Extension F16 (fibroblast + atrialCM); Below-detection 12 · F4 remains Below-detection (no fifth non-Below class) · nominal p (boxed * <0.05; dot <0.10); not FDR discovery · LA-only

A 16 NCP families × 9 LA co part ents

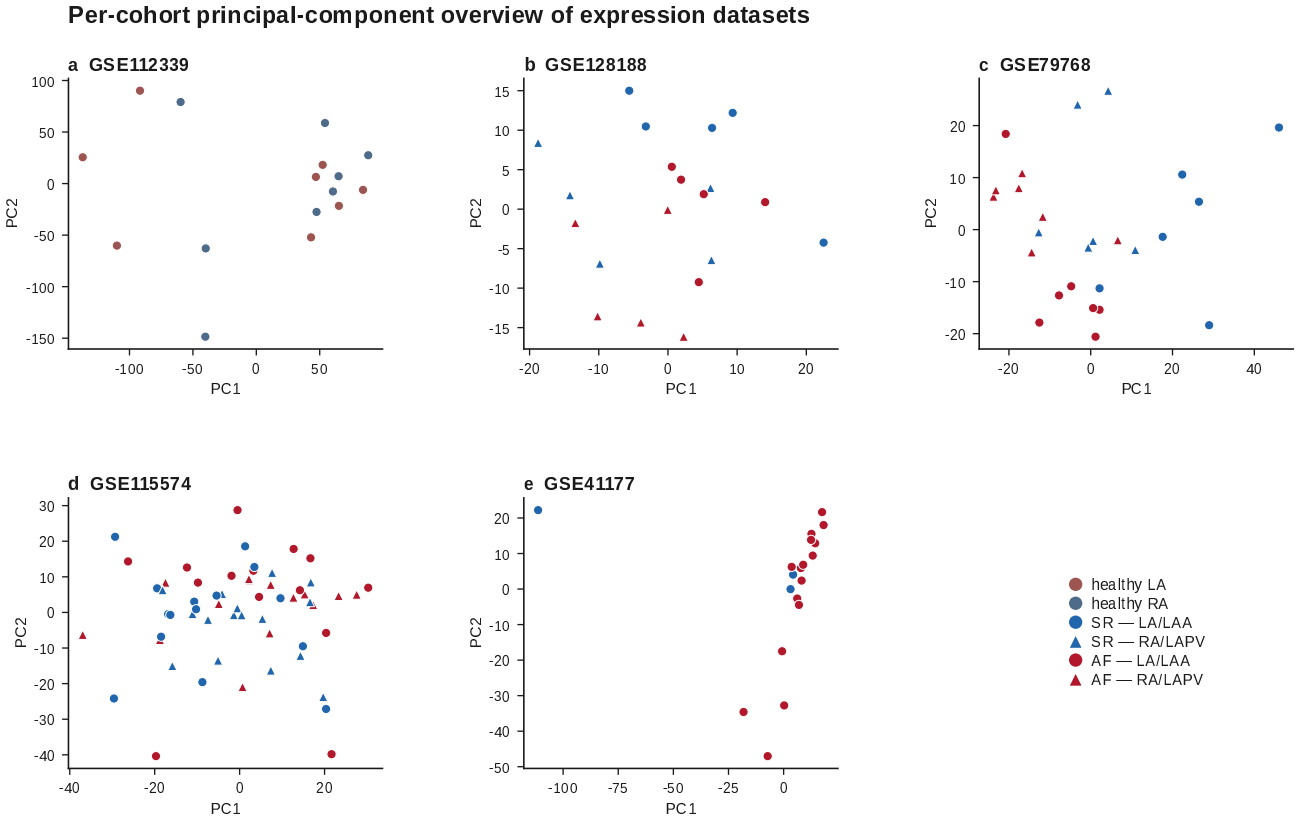


B 30 NCP programs × 9 LA co part ents



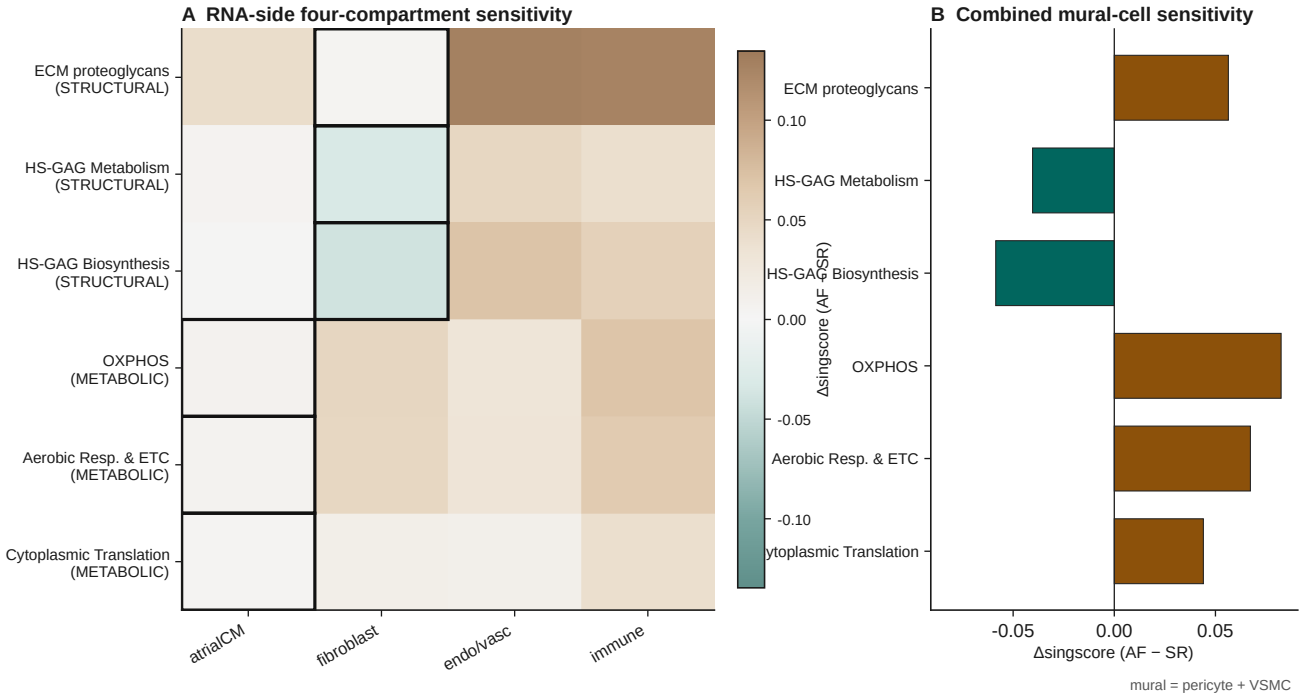
C Carrier-relative classes (nominal permutation; LA-only; not FDR-controlled)

Family	Class	Supported compartment(s)
F1	Below-detection	
F2	Below-detection	
F3	Below-detection	
F4	Below-detection	
F5	Below-detection	
F6	Below-detection	
F7	Below-detection	
F8	Below-detection	
F12	Below-detection	
F13	Below-detection	
F14	Below-detection	
F9	Below-detection	
F10	Fidelity	fibroblast
F11	Relay	adipocyte
F15	Relay	endothelial
F16	Extension	atrialCM,fibroblast

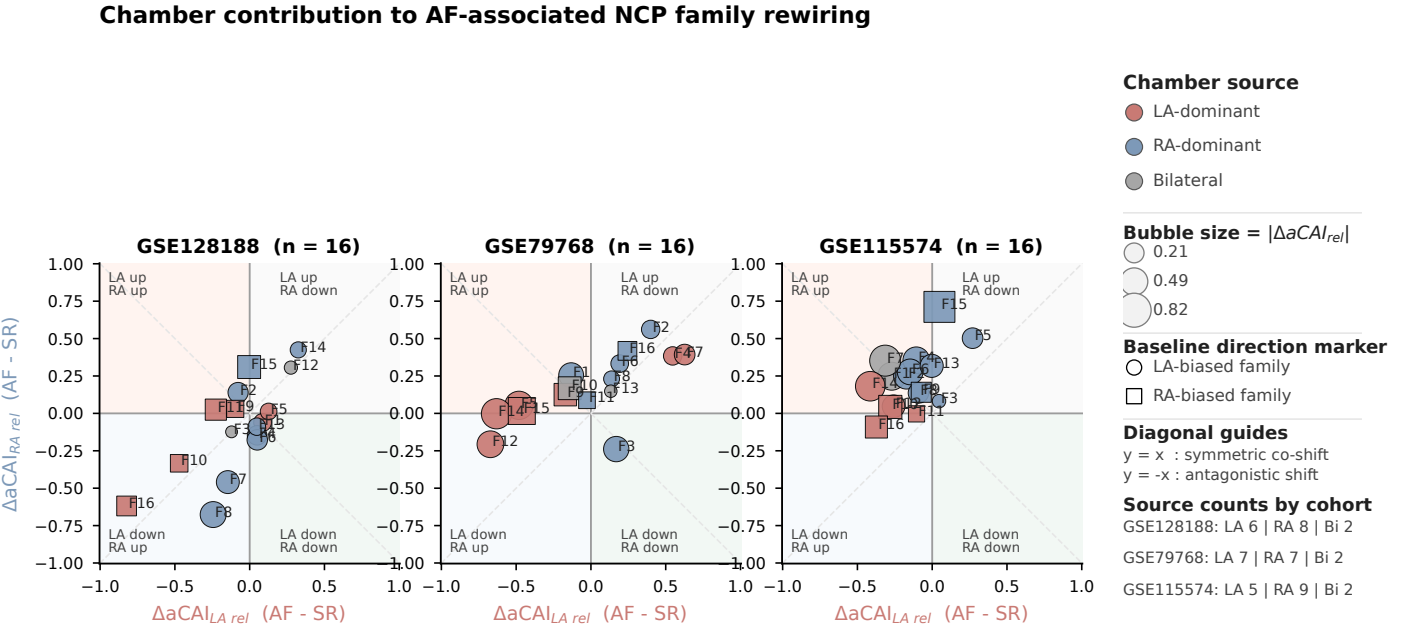

Per-cohort principal-component overview of expression datasets
a GSE112339
b GSE128188
c GSE79768
d GSE115574
e GSE41177
PC1
PC2
healthy LA
healthy RA
SR — LA/LAA
SR — RA/LAPV
AF — LA/LAA
AF — RA/LAPV
QC and sample-context visualization only; not an independent AF-remodeling test. Coordinates from frozen results/figS4_per_cohort_pca.tsv (no PCA recomputation).

GSE238242 RNA-side and mural-cell sensitivity

Supplementary sensitivity only · not a Path A gate · not an independent main-text evidence layer · does not alter Figure 3 localization classes or Figure 5 support counts · thick boxes mark locked expected RNA-side cells



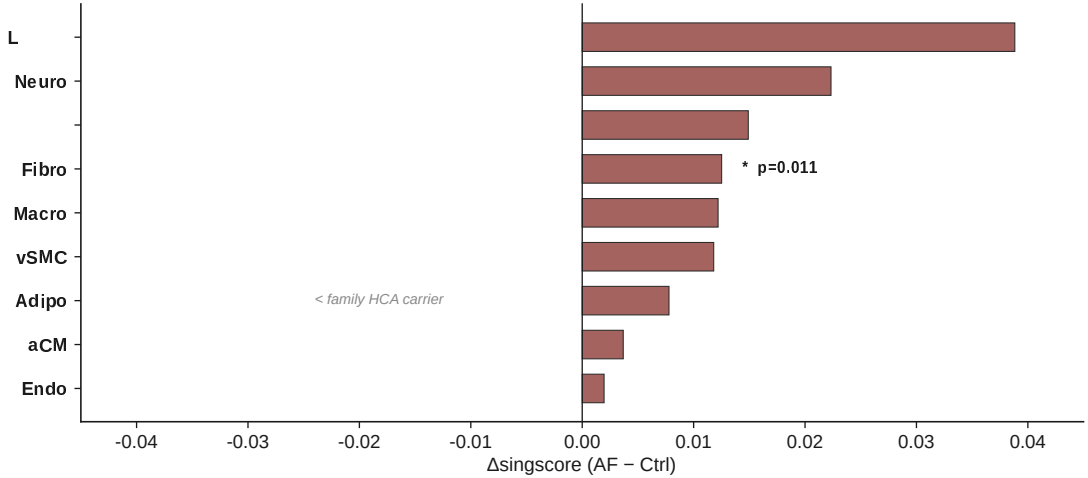
Frozen GSE238242 LAA snRNA pseudobulk Δsingscore (AF − SR). No expected RNA-side cell reaches moderate/strong support under the locked support-gate rule.



Each point is one family in one cohort. x = ΔaCAIrel_LA, y = ΔaCAIrel_RA (AF - SR). Point colour denotes the dominant chamber contribution (|LA| ≥ 1.25|RA|: LA-dominant

F4 BCAA constituent-program fibroblast support

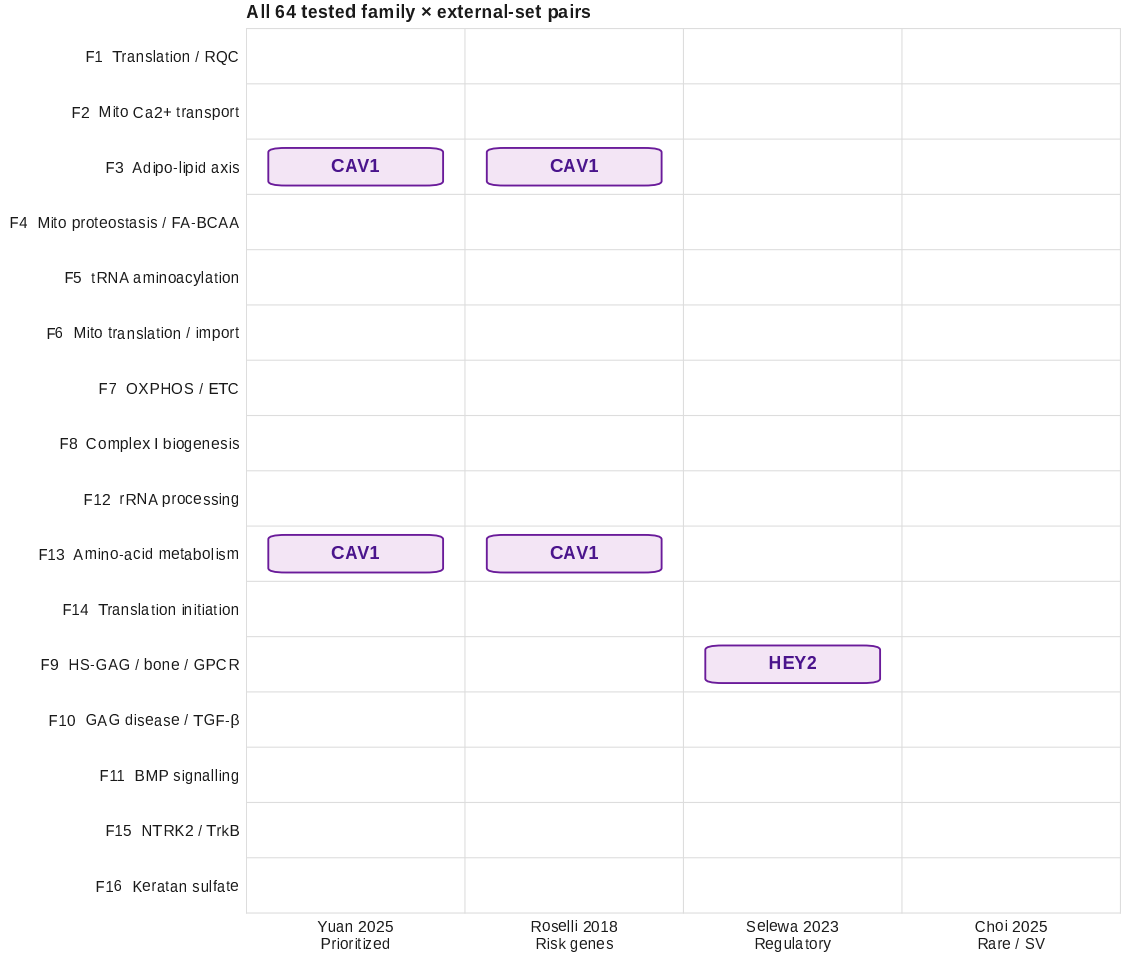
Program-level example only · parent F4 family remains Below-detection · not a fifth non-Below family · not FDR-controlled discovery · does not alter Figure 3 family classes or Figure 5 support counts



Corrected exact-reference singscore: BCAA constituent program (REACTOME branched-chain amino-acid catabolism) shows nominal fibroblast support (Δ=0.0125, p_perm=0.0115; n_ctrl=16, n_af=18). Family-level F4 class = Below-detection. HCA LA carrier prior for F4 = Adipo. Color encodes direction only (red AF>Ctrl; blue AF<Ctrl), not fidelity class.

Full NCP family overlap with external AF gene sets

16 families × 4 external sets · 5 nonzero cells · genes CAV1, HEY2 only · 0/64 BH-FDR-significant enrichments · universe N = 14,635 · member overlap ≠ family enrichment · blanks = tested zero overlap · not a Path A gate



Exact nonzero cells: F3×Yuan CAV1; F3×Roselli CAV1; F13×Yuan CAV1; F13×Roselli CAV1; F9×Selewa HEY2. Choi column tested with zero overlap. Repeated CAV1 cells are not independent genetic findings.